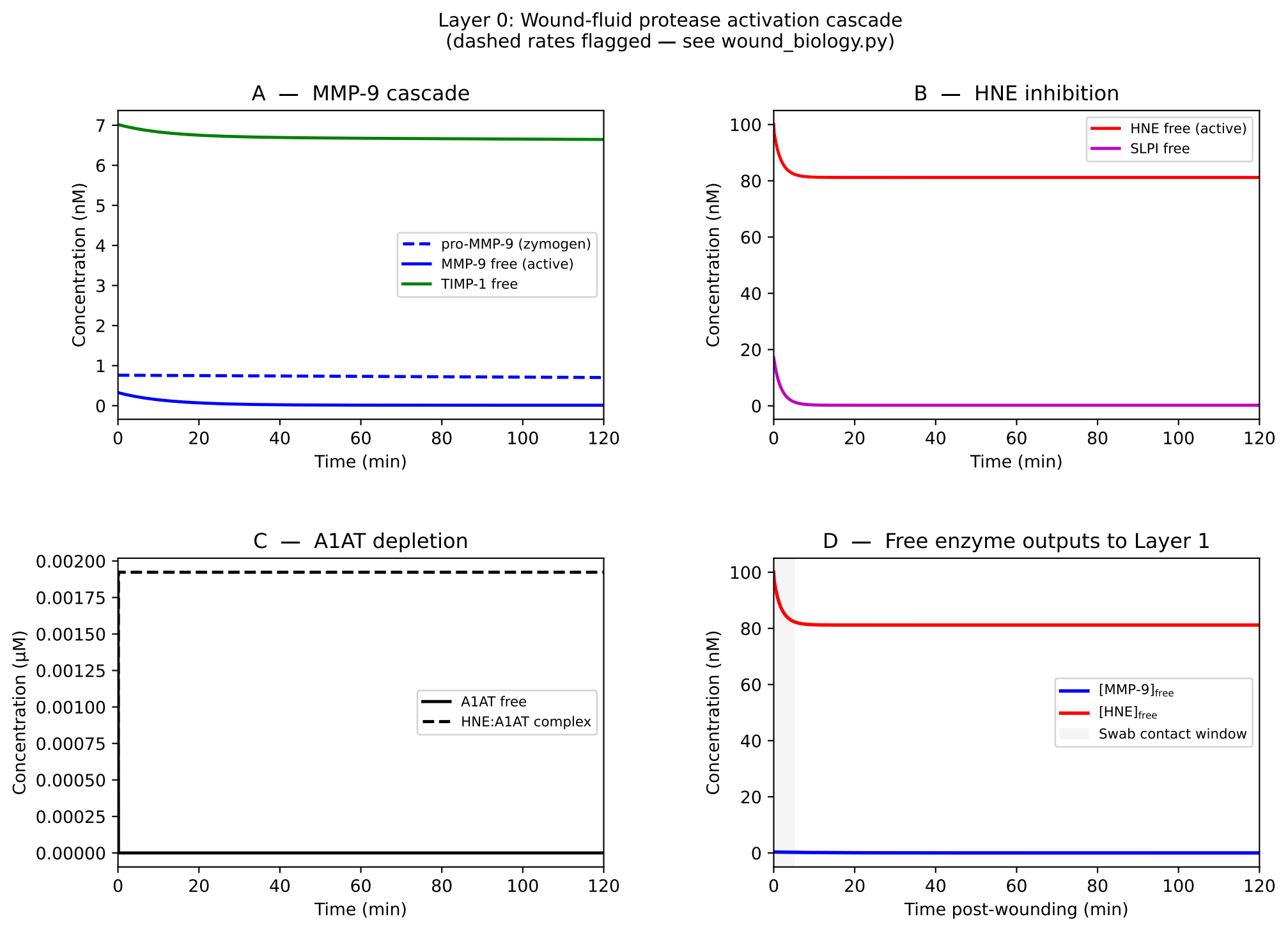
Layer 0: Wound-fluid protease activation cascade
(dashed rates flagged — see wound_biology.py)
A — MMP-9 cascade
Concentration (nM)
Time (min)
pro-MMP-9 (zymogen)
MMP-9 free (active)
TIMP-1 free
B — HNE inhibition
Concentration (nM)
Time (min)
HNE free (active)
SLPI free
C — A1AT depletion
Concentration (μM)
Time (min)
A1AT free
HNE:A1AT complex
D — Free enzyme outputs to Layer 1
Concentration (nM)
Time post-wounding (min)
[MMP-9]free
[HNE]free
Swab contact window